Quantum Computing

Part 10 - Industry, Academia & Claims: Applications & Benchmarking Claims

Compiled by the Spaced Research Ledger

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This part records claims that quantum computers have done something useful, together with the rebuttals those claims attracted. The two belong in the same document. A quantum advantage claim and its classical refutation are usually separated by months and reported by different outlets, and reading either alone gives a false picture.

The pattern repeats often enough to state plainly. A company announces that its machine solved a problem no classical computer could. Within weeks or months, a research group reproduces the result on conventional hardware, sometimes on a laptop. The claim is then defended, narrowed, or quietly dropped. Sections 1 and 2 should be read as a pair.

One distinction separates the strong claims from the weak ones. A complexity-theoretic claim argues that no classical algorithm could match the result, and rests on mathematical proof. An empirical claim reports only that known classical methods became unreliable in some regime, which allows a better classical method to appear later.

Section 1 covers advantage claims, Section 2 the rebuttals, Section 3 benchmark metrics, and Sections 4 through 7 the application areas: chemistry, optimization, machine learning and finance.

1. Quantum Advantage and Utility Claims

The 2026 claims are more carefully argued than their predecessors, and the companies making them are increasingly explicit about which kind of claim they are making.

Current State

In July 2026 IBM coordinated three preprint announcements, with the University of Chicago, Qedma and Algorithmiq [1]. They were presented together as evidence that quantum computing had entered an era of quantum advantage. The three papers carry different evidentiary weight, which the coordinated presentation tends to obscure.

The strongest of the three is the University of Chicago paper. It makes an explicit advantage claim backed by complexity-theoretic hardness arguments and a device-dependent fidelity certificate [1]. The Qedma and Algorithmiq papers make empirical claims about regimes where tested classical methods become unreliable.

The claim itself has a stated limit. IBM’s collaboration claims a mathematical proof that no classical computer can generate the specific sample distribution its circuit produced within a reasonable time [2]. The proof covers at least one such sample set and cannot rule out an unknown classical shortcut.

Earlier claims set the pattern. D-Wave's 2025 Science paper reported a quantum supremacy demonstration simulating spin glass dynamics across three lattice geometries on its Advantage2 annealer [3]. Google's original 2019 random circuit sampling claim drew intense scrutiny, and within months classical simulation had narrowed, though not closed, the gap [4]. Google and the University of Science and Technology of China have since repeated the experiment with larger systems and deeper circuits.

Quantinuum has demonstrated random circuit sampling on a trapped-ion system with fewer qubits but higher connectivity and lower error rates [4]. Its results are comparable to the superconducting experiments, reached through a different architecture.

2. Classical Simulation Rebuttals

Each entry here is a specific challenge to a specific claim in Section 1. One of them was answered by a laptop.

Current State

The most direct rebuttal came for D-Wave. Flatiron Institute physicists and collaborators matched D-Wave's published spin glass outputs across all three lattice geometries [3]. Their classical algorithm used three-dimensional tensor networks combined with a belief propagation method first published in 1982. It also agreed with analytically known answers on smaller instances.

D-Wave issued a formal rebuttal in May 2026 [5]. The company argues the classical method does not compute the full-state data profiles and higher-order observables its processor produced. It also does not attempt the most complex non-planar lattice topologies from the original paper. Both positions are recorded here because neither has withdrawn.

IBM's earlier utility claim met the same treatment. Its 2023 Nature paper on kicked Ising dynamics on a 127-qubit processor was challenged by several groups, including Caltech, whose classical methods reproduced the results substantially faster [6]. Those methods now serve as benchmarking tools for validating later utility claims, which is an unusually productive outcome for a refutation.

A July 2026 preprint challenges the certification method itself, proposing a frozen-tree algorithm that questions whether sample-only ensemble statistics can certify random circuit sampling at all [1]. It argues the algorithm reproduces the relevant ensemble-level statistics without simulating the circuit's full output distribution.

A 2026 benchmarking study reached a broader negative conclusion. On current noisy hardware, runtime-based quantum advantage has not been demonstrated under experimentally grounded metrics [7]. A restricted implementation of Simon's problem ran roughly two orders of magnitude slower than a tuned classical baseline. A claimed runtime advantage for one hybrid algorithm was not observed under more comprehensive testing.

3. Benchmark Suites and Metrics

Benchmarks are how any of the above gets compared. The recurring weakness in the established metrics is that most of them ignore speed.

Recent Developments

An August 2026 preprint formalizes a whole-system speed benchmark for layered, hardware-aware circuits on utility-scale processors, binding speed to independently verified layer-fidelity quality envelopes [8].

A June 2026 preprint introduces Clifford Volume and Free Fermion Volume as scalable volumetric benchmarks [9]. Both are classically simulable, which means their results can be verified rather than assumed, and both were validated experimentally on Quantinuum's H2-1 system, reaching a Clifford Volume of 34.

Current State

Quantum Volume, introduced by IBM in 2019, is a single-number benchmark that runs random square model circuits [10]. It folds qubit count, gate fidelity, connectivity, crosstalk, calibration and compiler quality into one score. Quantinuum's trapped-ion systems hold the highest reported values, reaching 2^{23} in May 2025 and a reported 2^{25} in September 2025 [10]. The company aims to increase the figure tenfold each year.

Circuit Layer Operations per Second, developed by IBM in 2021, measures execution speed by combining circuit runtime with real-time and near-time classical computation [11]. It was originally envisioned alongside qubit count and Quantum Volume as one of three core comparison metrics.

Algorithmic Qubits aims to measure the useful computational qubits a system has for practical algorithms [12]. Like Quantum Volume, it does not account for execution speed, which is the gap the recent work in this section addresses.

Two community efforts sit above the individual metrics. The QED-C suite is an open-source, evolving set of more than 20 application-based benchmarks, including phase estimation and Hamiltonian simulation, measuring execution quality, runtime cost and resource requirements [13]. Its architecture was updated in 2025 for modularity, integrating a low-level benchmark suite and NVIDIA's CUDA-Q for distributed simulation [14]. Metriq is a free collaborative platform supported by the Unitary Fund that aggregates benchmarking results from papers and news, and its framework integrates the QED-C benchmarks alongside device-level protocols [15].

4. Chemistry and Materials Simulation

Chemistry is the application most experts expect to benefit first. The claims below are correspondingly more measured than those in Sections 6 and 7.

Current State

Qedma researchers used an IBM machine to simulate a quantum magnet in a July 2026 preprint [2]. This is an empirical claim about a regime where classical methods reportedly become unreliable, and is deliberately weaker than the complexity-theoretic claim in the companion paper described in Section 1.

Quantinuum published a paper, now appearing in Nature, exploring a chemistry-relevant system at scales the company describes as frustrating classical computation [16]. It frames this as evidence that quantum computing is matching or exceeding classical high-performance computing in this domain.

Two applied collaborations are worth recording. Investigators highlighted by St. Jude Children's Research Hospital reported that augmenting machine-learning drug discovery with quantum computing outperformed comparable classical-only models in identifying compounds against a KRAS target [17]. The work included experimental validation rather than simulation alone, which is rare in this area. Pasqal and Qubit Pharmaceuticals applied hybrid quantum methods to protein hydration and ligand-protein binding, focused on placing water molecules accurately within protein cavities [17].

One claim carries a caveat. A 2026 preprint claims a parallel method enables quantum chemistry simulations at 200-qubit scale on accelerated computing platforms, reporting results said to surpass classical benchmarks for ruthenium catalysts [18]. It has not been peer-reviewed.

5. Optimization Applications

Optimization attracts the most commercial interest and produces the least verified advantage. The most useful recent contribution here is about how to measure the claims rather than a claim itself.

Recent Developments

Two Fraunhofer publications from August 2026 propose methods for assessing potential quantum advantage under more realistic physical conditions and across increasing problem sizes [19]. This is a contribution to benchmarking rigour rather than a new application claim.

Current State

The strongest recent result is hybrid. A July 2026 preprint by JPMorgan Chase, AWS, Quantinuum and collaborators reports a hybrid quantum-classical portfolio optimization workflow on Quantinuum's 98-qubit Helios hardware [20]. It consistently outperformed a standalone quantum algorithm on real market data from four major financial indices. The comparison is against another quantum method, not against the best classical one.

A February 2026 preprint testing a constraint-preserving formulation for stock selection reported a Sharpe ratio of 1.81 [21]. The monthly walk-forward backtest on 10 US equities gave 1.31 for simulated annealing and 0.98 for a standard classical method. The sample is small and the source confidence low.

A September 2025 Fraunhofer benchmark compared five methods on realistic portfolio instances [21]. The quantum approximate optimization algorithm reached 97.3% of the exact classical solution quality in certain parameter ranges, for small and medium problems. Hybrid strategies showed the most promise for larger ones. Reaching 97.3% of a classical answer is not an advantage over it.

Two supporting items complete the picture. The Quantum Optimization Benchmarking Library is a 2025 preprint establishing a standard suite for evaluating quantum optimization algorithms [22]. D-Wave specializes in quantum annealing for combinatorial optimization, with systems reportedly used by enterprises including NASA and Volkswagen, though these are vendor-reported deployments rather than peer-reviewed demonstrations [23].

6. Machine Learning Applications

The theory in this section is stronger than the practice. One paper establishes when a quantum advantage could be inherited at all, which is more than most application claims can offer.

Recent Developments

A July 2026 paper derives the conditions under which quantum machine learning can inherit computational advantages from general quantum algorithms for supervised learning [24]. Establishing when an advantage is even possible is a firmer foundation than any individual benchmark.

A July 2026 preprint by WISER and E.ON reports hybrid benchmarks for multi-output electricity demand forecasting on IBM hardware with more than 100 qubits [25]. One model achieved a 40.37% reduction in mean absolute error against a classical baseline on real hardware, and 62.01% in simulation. The gap between the hardware and simulation figures is itself the finding.

Current State

Quantum kernel methods and quantum neural networks are being explored for fraud detection, to identify nonlinear patterns in high-dimensional data [26]. Practical advantage there remains out of reach, with most researchers expecting meaningful results only from larger and more accurate processors.

One widely reported result needs its qualification attached. IonQ and Ansys reported in early 2025 that a hybrid algorithm delivered up to 12% faster processing than classical methods on a blood pump simulation [23]. The full-scale speedup was achieved by classically simulating the quantum algorithm rather than running it, and IonQ hardware was used only to validate smaller instances.

7. Finance Applications

Banks are paying for access to quantum computers. As of the most recent assessment here, none of them has obtained an advantage from it.

Recent Developments

ORIENTOM and softwareQ signed a memorandum of understanding in August 2026 to explore quantum and quantum-inspired computing for financial services, banking and insurance [27]. Named use cases include derivatives pricing, portfolio optimization, risk management and machine learning. This is an exploratory partnership rather than a benchmarked result.

Current State

The summary judgment is direct. As of March 2026, no quantum advantage had been demonstrated for any real-world financial application, despite banks paying for quantum computing access in anticipation of future advantage [26].

Commercial activity continues regardless. Rigetti announced a partnership with Algorithmiq in February 2026 targeting quantum machine learning for fraud detection, among the first commercial collaborations aimed at that use case [26]. IBM has the deepest financial-sector engagement of any vendor through its quantum network, which includes JPMorgan Chase, Goldman Sachs and Barclays [26]. Its finance module offers pre-built circuits for portfolio optimization, option pricing and risk analysis.

Regulators are watching the same gap. The UK Financial Conduct Authority highlighted optimization, machine learning and stochastic modelling as material domains for financial-services quantum applications [28]. It stressed benchmarking, explainability, validation and resilience as prerequisites for adoption, rather than confirming any current advantage.

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

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